

Machine Learning Theory: Miniproject

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1 Chosen topic

Can LLMs perform statistical analysis? Preparation of a prompt (with results) for, e.g., GPT-4, which can provide ready-to-use code to conduct, for example, one of the above experiment. Chosen experiment: Characterization of error on the training and test sets as a function of regularization strength in linear regression / logistic regression / SVM (note: the regularization coefficient can be selected on an exponential scale, e.g., 2^{-10} , 2^{-9} , $\dots$, 2^9 , 2^{10} , etc.).

2 Preparation of a prompt

I need to conduct an experiment to characterize the error on the training and test sets as a function of regularization strength for linear regression, logistic regression, and SVM. Can you provide Python code using scikit-learn that does the following:

1. Loads the California housing dataset for linear regression.
2. Loads the Breast Cancer dataset for logistic regression and SVM.
3. Splits each dataset into training and test sets.
4. For linear regression:
 - 4.1. Trains a linear regression with L2 regularization model with alpha ranging from 2^{-10} to 2^{10} .
 - 4.2. Plots the training and test errors as a function of regularization strength.
 - 4.3. Prints the optimal regularization strength based on the test error.
5. For logistic regression:
 - 5.1. Trains a logistic regression model with L2 regularization with C (inverse of regularization strength) ranging from 2^{-10} to 2^{10} .
 - 5.2. Plots the training and test errors as a function of regularization strength.
 - 5.3. Prints the optimal regularization strength based on the test error.
6. For SVM:
 - 6.1. Trains an SVM model with L2 regularization with C (inverse of regularization strength) ranging from 2^{-10} to 2^{10} .
 - 6.2. Plots the training and test errors as a function of regularization strength.
 - 6.3. Prints the optimal regularization strength based on the test error.

This prompt was given to GPT-4 and the results are uploaded to [Github](#) repository. The dataset used for linear regression is California Housing dataset as it contains continuous target values representing house prices, making it suitable for linear regression analysis. For logistic regression and SVM Breast Cancer dataset was used as it contains binary target values indicating the presence or absence of cancer, making it suitable for classification analysis with logistic regression and SVM.

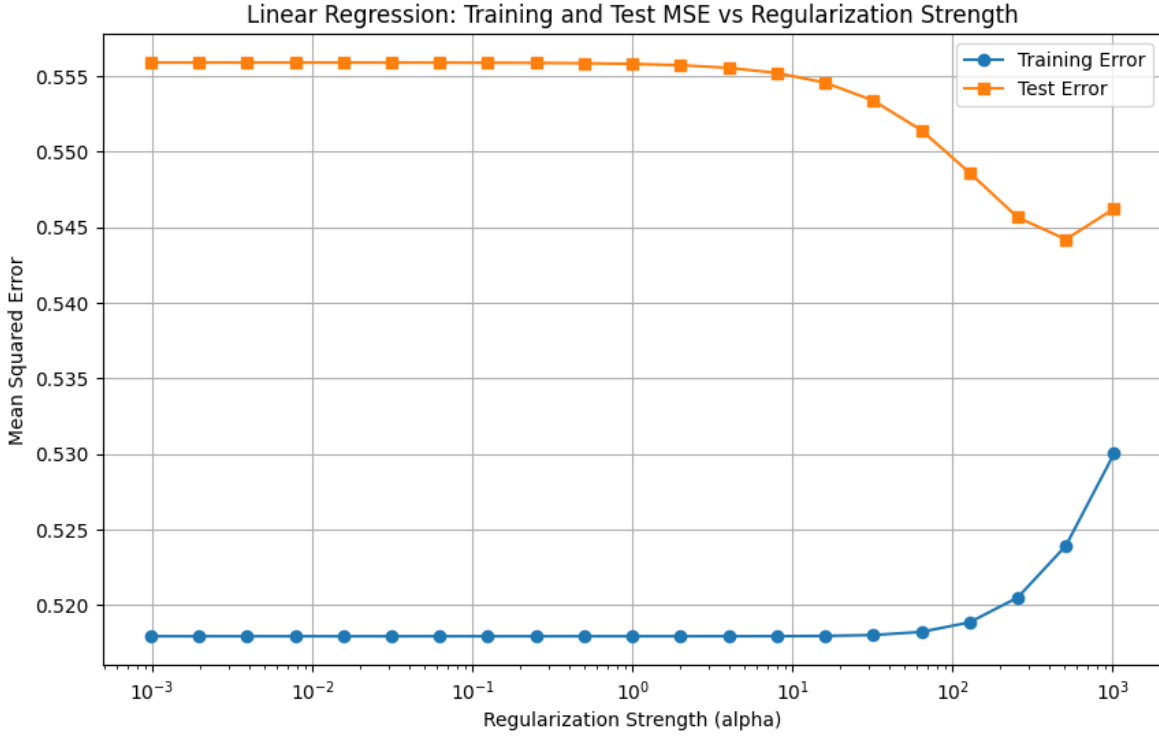


Figure 1: Linear Regression

3 Results

3.1 Linear regression

The plot in Figure 1 shows that the training error remains relatively low and stable for small regularization strengths (alpha), but it increases significantly for larger alpha values. The test error is also quite stable across a wide range of regularization strengths, decreasing slightly for moderate alpha values, but it increases again for very large alpha values. The optimal regularization strength based on test MSE: 512.0.

To conclude, low and moderate regularization strengths help in reducing overfitting and yield lower test errors, but very high regularization strengths lead to underfitting, increasing both training and test errors.

3.2 Logistic regression

The second plot (Figure 2) presents that the training error decreases as the inverse regularization strength (C) increases, indicating that lower regularization (higher C) allows the model to fit the training data better. The test error remains relatively stable across the range of C values but shows minor fluctuations at the beginning. The optimal regularization strength (C) based on test error: 0.0009765625.

To sum up, higher values of C (lower regularization) tend to improve the model's performance on the training set but do not significantly impact the test set performance, indicating a balance between bias and variance. Logistic regression is less sensitive to changes in regularization strength compared to linear regression, maintaining stable test error across a wide range of C values.

3.3 SVM

The last plot (Figure 3) shows that the training error decreases sharply with an increase in C up to a certain point and then stabilizes, the test error follows a similar pattern, with a sharp decrease initially and then a stabilization. The optimal regularization strength (C) based on test error: 64.0.



Figure 2: Logistic Regression

In conclusion, moderate values of C are optimal for minimizing the test error. SVM benefits from some regularization to prevent overfitting, but very low regularization (high C) can lead to overfitting on the training data

4 Conclusions

The choice of regularization strength (alpha for linear regression and C for logistic regression and SVM) significantly impacts model performance. Proper tuning is essential to balance bias and variance. Logistic regression is more robust to changes in regularization strength compared to linear regression and SVM. While reducing training error is relatively straightforward by lowering regularization, minimizing test error requires careful tuning to avoid overfitting and underfitting.

When it comes to the GPT-4 ability to perform statistical analysis, it was able to generate accurate and executable Python code for performing the statistical analysis tasks. The code included loading datasets, splitting them into training and test sets, training models with varying regularization strengths, and plotting the results. The generated code followed the provided instructions, ensuring that all specified tasks (e.g., training models, plotting errors, printing optimal regularization strengths) were completed correctly.

The models were trained correctly with varying regularization strengths, and the results were as expected, demonstrating GPT-4's capability to handle machine learning tasks effectively. The plots generated for training and test errors as a function of regularization strength were clear, well-labeled, and accurately represented the data. This indicates GPT-4's proficiency in generating code for data visualization.

In summary, GPT-4 has demonstrated strong capabilities in performing statistical analysis and generating machine learning code. Its ability to accurately interpret and execute instructions makes it a powerful tool for both educational and practical applications in data science and machine learning.

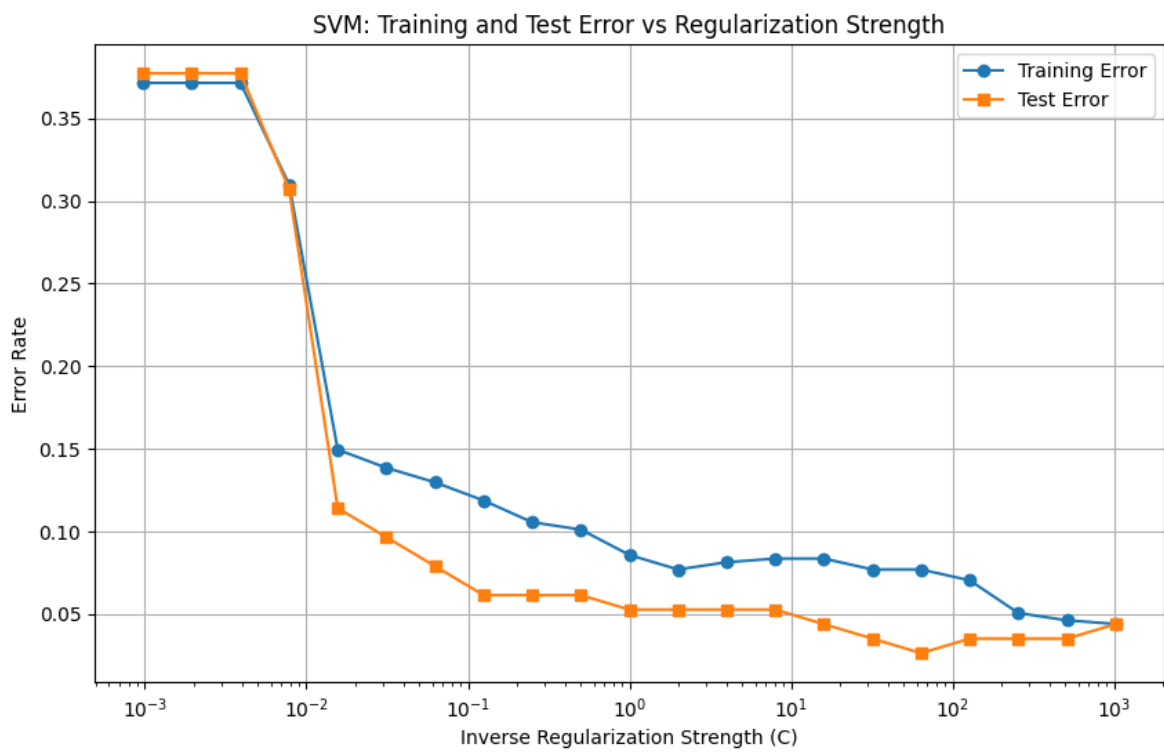


Figure 3: SVM